

1 1

This question extends the Skeleton Program by allowing a user to simulate a light breeze.

When a light breeze occurs, all pheromones are moved one position to the right, as shown in **Figure 1**.

Figure 1

Before Breeze:					After Breeze:				
				P					
			P					P	
	P	P					P	P	
	N					N			

Where N is a Nest and P is a Pheromone within a cell.

No other elements of the simulation should be changed.

Pheromones that are moved should have the same strength before and after they move.

What you need to do

Task 1

Create a new method in the `Simulation` class called `LightBreeze` that moves all Pheromones as described as described.

Task 2

Add a new menu option, '6. Light Breeze', within the main menu to allow the user to access this feature. Then the `LightBreeze` method is called.

Task 3

Test that the changes you have made work:

- run the Skeleton Program
- enter 1
- enter 5 then 5
- enter 1
- enter 6
- enter 1

Evidence that you need to provide

Include the following evidence in your Electronic Answer Document.

1 1 . 1

Your PROGRAM SOURCE CODE for the new method `LightBreeze` and any amended methods.

[4 marks]

1 1 . 2

SCREEN CAPTURE(S) showing the results of the requested test.

[1 marks]

5

